



Learning Agility Measurement Models in Technology-Driven Learning Environments: A Systematic Literature Review

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ABSTRACT

Purpose - This study systematically examines how learning agility has been measured in recent literature by identifying dominant measurement models, dimensional structures, and psychometric approaches across contexts.

Methods - This study employed a Systematic Literature Review (SLR) guided by the PRISMA 2020 statement. A structured search was conducted in the Scopus database using predefined keywords. The selection process followed identification, screening, eligibility, and inclusion stages based on criteria such as publication year (2020–2025), document type (peer-reviewed journal articles), language (English), and full-text accessibility. A total of 24 articles were included and appraised using the Mixed Methods Appraisal Tool (MMAT) and analyzed through systematic synthesis.

Findings - Learning agility is predominantly measured using self-report Likert-scale instruments, with validation through exploratory factor analysis (EFA) and confirmatory factor analysis (CFA). The four-dimensional model—cognitive, people, change, and results agility—remains the most widely used, with variations influenced by context. Analytical techniques are dominated by Structural Equation Modeling (SEM) and Partial Least Squares SEM (PLS-SEM), with learning agility often positioned as a mediating variable.

Research Implications - Studies show overreliance on single-source data and limited use of behavioral, longitudinal, and multimethod approaches, indicating the need for more comprehensive, context-sensitive, and technology-integrated measurement models.

Originality - This study synthesizes learning agility measurement approaches across contexts, clarifies dominant models, and identifies methodological gaps, highlighting the need for more robust and diversified measurement strategies.

INTRODUCTION

In recent years, the integration of digital technologies, such as e-learning platforms, learning management systems (LMS), and artificial intelligence (AI)-based tools, has transformed the way individuals develop and demonstrate learning agility. In technology-driven environments, learning agility is viewed as a psychological construct and a capability that interacts with digital systems, adaptive learning technologies, and human–computer interaction processes. Therefore,

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understanding how learning agility is measured is increasingly important in the context of digital and technology-enhanced learning.

Over the past five years, rapid changes in the world of work and education have made learning agility a skill that is gaining increasing attention across various professional fields. While previously discussed primarily in the context of leadership and talent management, the concept has now expanded to include universities, the teaching profession, public institutions, and technology-driven workplaces ([Chang & Octoyuda, 2024](#); [Reeves & Whitaker, 2020](#)). The ability to quickly absorb knowledge, adapt approaches, and shift mindsets to meet the demands of a situation has made learning agility a key indicator of an individual's readiness to face the era of digital transformation. Recent empirical findings suggest that individuals with high learning agility tend to demonstrate greater creativity, openness to new experiences, and effectiveness in solving complex problems under conditions of uncertainty. These attributes make learning agility a critical asset in modern organizations ([Goldsmith, 2020](#)). However, the increasing attention to learning agility has given rise to diverse definitions and variations in how it is measured. Unlike established psychological constructs, this concept has evolved through many theoretical and methodological approaches, leading to inconsistent research findings and fragmentation in the literature.

One of the most widely recognized frameworks in learning agility research is the model proposed by Lombardo and Eichinger, which conceptualizes learning agility as a multi-dimensional construct. This framework typically includes several core dimensions, such as cognitive, people, results, mental, and change agility, which together capture an individual's ability to learn from experience and adapt effectively across situations. Despite its widespread use, recent studies have questioned the adequacy of this model in addressing contemporary organizational demands. For example, [Boulard-van Dam et al. \(2022\)](#) introduced the Leadership Learning Agility Scale (LLAS), which consists of three dimensions: Developing Leadership, Seeking Feedback, and Developing Systematically. This alternative model is considered to be more aligned with modern leadership practices and evolving workplace expectations.

In addition to the diversity of models, several studies have identified conceptual overlaps between learning agility and related constructs, such as openness to experience, adaptability, and proactivity. [Smith & Watkins \(2024\)](#) showed that some measurement instruments contain items that closely resemble these constructs, raising concerns regarding construct validity and dimensional distinctiveness. Furthermore, inconsistencies in reliability and factor structures have been observed when instruments are applied across different cultural contexts, indicating potential limitations in generalizability.

At the regional level, variations in the measurement approaches are increasingly evident. [Megawaty et al. \(2025\)](#) for instance, adapted a learning agility instrument for the Indonesian civil service and found that culturally sensitive modifications were necessary to improve its accuracy. In educational contexts, learning agility is often associated with reflective practice rather than rapid behavioral change, which is more commonly emphasized in industrial settings. These findings suggest that measurement instruments developed in Western contexts may not be directly applicable to Asian or public sector environments without appropriate adaptation.

Although several literature reviews on learning agility have been conducted, most have focused on specific domains such as human resource development, organizational behavior, or work engagement. For example, [Boulard-van Dam et al., 2022](#) examined learning agility within the context of human resource development, while ([Sapbamrer & Thammachai, 2021](#)) reviewed models and measurement approaches more broadly. Similarly, ([Smith & Watkins, 2024](#)) provided a critical review of measurement instruments, ([Hidayat et al., 2025](#)) explored the role of learning agility in performance outcomes, and ([Nareswari & Astuti, 2025](#)) conducted a scientometric analysis linking learning agility with work engagement. However, these studies differ in scope, methodological approach, and analytical focus and do not provide a fully integrated and up-to-date synthesis specifically addressing the diversity of measurement models and their psychometric characteristics in the recent literature.

Given these limitations, a more systematic and comprehensive synthesis is needed to map the current landscape of learning agility measurements. As highlighted by [Snyder \(2019\)](#), systematic

literature reviews offer a structured and transparent approach to identifying patterns, methodological trends, and research gaps within a field. Therefore, this study adopts a Systematic Literature Review (SLR) approach, following PRISMA guidelines, to systematically identify, evaluate, and synthesize contemporary empirical and conceptual studies on learning agility measurement. This SLR focuses on conceptual models, measurement instruments, psychometric properties, and methodological issues across studies published between 2020 and 2025. This study aims to address inconsistencies in the literature and provide a clearer understanding of how learning agility is operationalized in different contexts.

The research question guiding this study is as follows:

RQ1. How is learning agility measured in existing empirical and conceptual studies, and which models, dimensions, and measurement frameworks are most commonly used?

Through this systematic mapping, this study is expected to contribute theoretically by providing a structured and integrative overview of learning agility measurement models. Practically, the findings are intended to support researchers and practitioners in selecting, adapting, and developing more valid and context-appropriate measurement instruments.

METHOD

Research Design

This study employs a Systematic Literature Review (SLR) design to provide a comprehensive mapping of models and instruments used to measure learning agility in recent scientific publications. The SLR approach was selected because of the rapid development of learning agility literature, which is dispersed across multiple disciplines such as psychology, management, education, and human resource development. The diversity of definitions, theoretical perspectives, and measurement instruments makes it difficult to compare the findings across studies. Therefore, an SLR approach was used to develop a systematic, transparent, and replicable synthesis.

The review process follows the PRISMA 2020 guidelines [Page et al. \(2021\)](#), which provide a structured framework consisting of identification, screening, eligibility, and inclusion stages. In addition, general principles of systematic literature reviews proposed by [Donthu et al. \(2021\)](#) and [Xiao & Watson \(2019\)](#) were used to guide the overall review process, including the formulation of research objectives and synthesis of findings. Through this design, this study aims to identify, compare, and evaluate the various instruments used to measure learning agility, as well as examine their conceptual and methodological consistency across studies.

Identification

In the identification stage, the researchers searched for articles in the Scopus database on November 11, 2025. The search was conducted by limiting the focus to the title, abstract, and keywords, using a combination of the terms "Learning Agility" AND "model" to ensure that the articles obtained were truly relevant to the topic. From this initial search, 135 articles were collected and recorded as an initial dataset for further processing in the filtering stage according to the PRISMA flow. The selection of Scopus as the primary source and the use of a title-abstract-keywords- based search strategy followed the literature identification standards commonly used in contemporary SLR methods.[\(Hidayat et al., 2025; Page et al., 2021\)](#).

This study focuses on the Scopus database because of its extensive coverage of high-quality peer-reviewed international publications and its standardized indexing system. Although the use of a single database may limit comprehensiveness, it ensures consistency and reliability of data selection. Future studies should incorporate additional databases, such as Web of Science, ERIC, and PsycINFO, to enhance the breadth of the analysis.

Table 1. The search string used for the systematic review process

Database	Search File	Search String
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Scopus	TITLE-ABS-KEY	TITLE-ABS-KEY ("learning agility" AND (measure* OR scale* OR instrument* OR model* OR assess* OR evaluat* OR validat* OR reliability OR dimension* OR framework))
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Screening

In the screening stage, a gradual screening process was conducted to ensure that the remaining articles were relevant and met the quality standards. The initial search identified 135 articles in the databases. The first screening was based on the publication year, with only articles published between 2020 and 2025 retained. Of the initial articles, 30 were eliminated because they were published before 2020, leaving 105 remaining articles. This year restriction was applied to ensure the review remained up-to-date with the latest literature, as recommended in modern SLR practice(Hidayat et al., 2025).

The next stage was the selection based on the document type. Articles that did not fall into journal categories such as book chapters, conference papers, reviews, books, notes, letters, and conference reviews were excluded. In this stage, 34 articles were removed based on document type criteria, resulting in 71 remaining articles. This step refers to the PRISMA recommendation, which emphasizes the use of peer-reviewed journal articles to maintain the quality of findings(Page et al., 2021). The screening then continued with language aspects, and only English-language articles were retained for the review. Articles written in other languages, such as Korean and Persian, were excluded to minimize interpretation bias and ensure consistency in data extraction, as commonly applied in systematic reviews following PRISMA guidelines (Page et al., 2021; Rethlefsen et al., 2021). After this stage, 69 articles remained.

The screening stage was then conducted by considering the full-text accessibility. Of the 69 articles that passed the language screening, 36 were excluded due to the lack of complete full-text availability, resulting in 33 articles that met the full-text screening criteria. Full-text availability was essential to enable in-depth analysis; however, inclusion was primarily based on relevance, methodological rigor, and completeness of reported data (Nareswari & Astuti, 2025). Of these 33 articles, only 24 were successfully retrieved in full-text format, while the remaining nine articles were not available for download. Therefore, 24 articles were included in the final analysis and used for in-depth synthesis.

Table 2. Inclusion criteria for article screening

Criterion	Inclusion	Exlusion
Timeline	Between 2020-2025	2019 and earlier
Document Type	Journal (research article)	Book Chapter, Conference paper, Review, Book, Note, Letter, Conference review
Language	English	Non-English
Full-text Access	Available	Not Available
Download Available	Downloaded	Not Downloaded

Included

In the inclusion stage, researchers determined the final list of articles eligible for analysis after In the inclusion stage, researchers determined the final list of articles eligible for analysis after completing the identification, screening, and eligibility assessment processes. Of the 33 articles that met the screening criteria, only 24 were successfully retrieved in the full-text format. Although the remaining nine articles were classified as open access, they were not available in full text. Therefore, only articles with fully accessible and retrievable full-text versions were included to ensure the accuracy and completeness of the data extraction.

Prior to final inclusion, the 24 selected articles underwent a quality appraisal process using the Mixed Methods Appraisal Tool (MMAT). Two independent reviewers conducted the assessment to ensure objectivity and consistency. The appraisal results indicated that all the included studies met the

required methodological quality standards, with a high level of agreement between the two reviewers, confirming the reliability of the evaluation process. Therefore, all articles were deemed suitable for inclusion in the final analysis, and the study proceeded to the results stage.

The inclusion of articles was not solely based on access status but primarily on their relevance to the research objectives and the completeness of the reported information required in the extraction table. All selected articles were further evaluated to ensure the accuracy and clarity of scientific reporting, in line with PRISMA principles that emphasize transparency in study selection (Page et al., 2021).

Furthermore, the final selection process was conducted systematically to ensure that only studies contributing significantly to the analysis of learning agility models, dimensions, and measurement approaches were included, as suggested by recent SLR research in learning management (Bouland-van Dam et al., 2022). Therefore, these 24 articles served as the primary basis for the synthesis and discussion in this review.

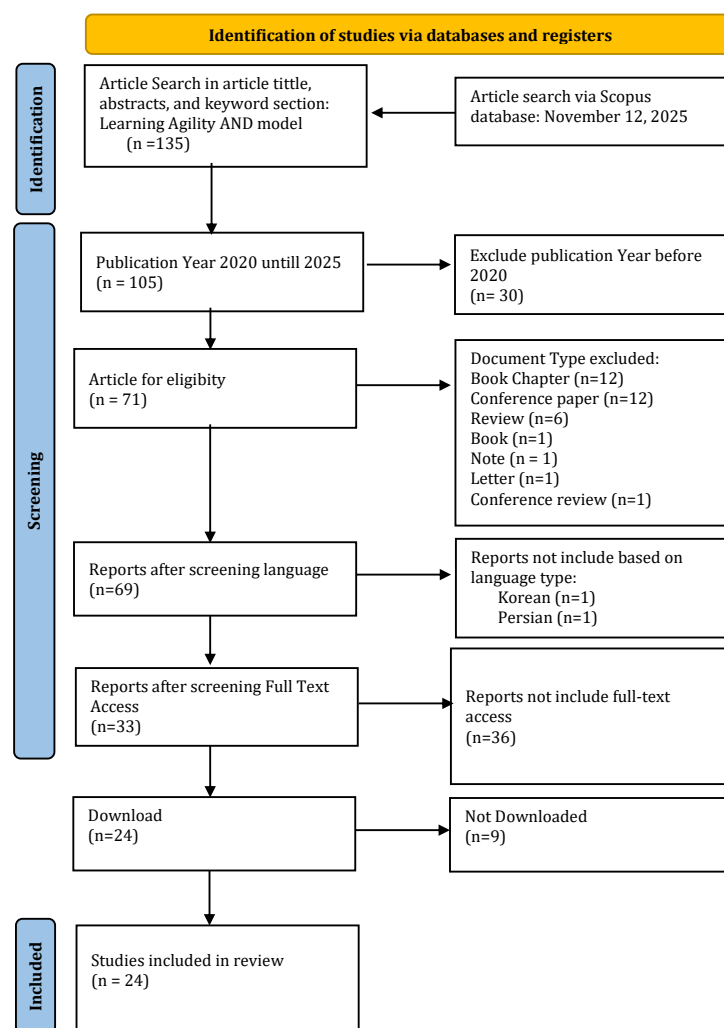


Figure 1. PRISMA diagram

RESULTS AND DISCUSSION

Results

Reference tracing analysis is a widely used approach to map and understand the development of a field of study, as it allows for the structured tracing of publication patterns and the dynamics of research themes. In this study, this method was applied to 24 articles discussing learning agility and related constructs in various contexts. Table 3 summarizes various studies addressing the topic of learning agility across a broad range of issues, including the work environment, employee performance, innovative behavior, leadership, education, mental health, and the application of digital technology. This diverse focus reflects the widespread use of learning agility as a key variable in assessing individual adaptability in organizations and educational institutions.

The findings from this reference search analysis provide a more comprehensive overview of the learning agility research landscape. The results indicate that scholarly attention is not solely focused on leadership development or organizational performance but also extends to education, the business world, and aspects of psychological well-being. Therefore, the reference search analysis complements the systematic review results, as it not only presents the content of each study but also demonstrates how the direction and thematic distribution of studies on learning agility have evolved over time.

Table 3. Data Extraction of Learning Agility Measurement Across Selected Studies

No	Author & Year	Instrument /Model	Dimensions of Learning Agility	Sample/ Context	Analytical Method	Psychometric Findings
1	(Li Chalermvo ngsavej, 2025)	Questionnaire -based learning agility model	Emotional intelligence, self-leadership, self-efficacy, achievement motivation	Undergraduate students (China)	CFA & SEM	Model validated through CFA
2	(Carmeli et al., 2021)	Conceptual learning agility framework	Psychological availability, generativity, reflective reframing	Inter-organizational relationships	Conceptual analysis	Theoretical model development
3	(Narajah et al., 2025)	Survey questionnaire	People agility, change agility, mental agility, result agility	Engineers in SMEs (Malaysia)	Descriptive statistics	Measurement applied empirically
4	(Wolor et al., 2025)	Structured questionnaire	Innovative behavior, adaptability, performance orientation	IT employees (Indonesia)	SEM	Structural model tested
5	(Grau García et al., 2025)	BERT-based topic modeling	Cognitive patterns, knowledge adaptation, learning themes	Scientific publications dataset	NLP & topic modeling	Model identified thematic clusters
6	(Jung & Lee, 2024)	Competency measurement scale	Learning adaptability, emotional competence, professional capability	Nurses	EFA & CFA	High reliability ($\alpha > 0.9$)

Table 3 (continued). Data Extraction of Learning Agility Measurement Across Selected Studies

No	Author & Year	Instrument /Model	Dimensions of Learning Agility	Sample/ Context	Analytical Method	Psychometric Findings
7	(Konstantinou & Miller, 2021)	Workshop-based model	Teamwork, communication, reflective learning	Employees	Qualitative analysis	Skills development identified
8	(Zykiene et al., 2021)	Composite index model	Adaptability, regional responsiveness, development capability	Regional economic data	Index analysis	Indicators statistically validated
9	(Ab Jalil et al., 2022)	Questionnaire model	Learning experience, curriculum adaptability, future readiness	Students	PLS-SEM	Model predictive relevance established
10	(Tripathi & Kalia, 2024)	DigiComp-based instrument	Digital competence, adaptive learning, skill development	Learners	SEM	Construct validity supported
11	(Patwardhan et al., 2022)	Organizational survey instrument	Learning culture, adaptability, innovation capability	Employees	SEM	Hypothesis supported
12	(Lian, 2022)	Academic questionnaire	Self-efficacy, academic motivation, adaptive learning	Students	SEM (SmartPLS)	Significant relationships confirmed
13	(Io & Hong, 2022)	Survey instrument	Employee engagement, organizational support, adaptability	Employees	Moderated regression	Model relationships significant
14	(Bouland-van Dam et al., 2022a)	LLAS scale	Developing leadership, feedback seeking, systematic development	Leaders	Scale validation	Valid and reliable scale
15	(Huang & Kou, 2025)	Mixed-method instrument	Resilience, academic buoyancy, psychological well-being	Students	SEM + interviews	Model supported qualitatively & quantitatively
16	(Mahmutoglu et al., 2025)	Survey questionnaire	Leadership influence, organizational commitment, adaptability	Teachers	Regression analysis	Statistical significance achieved
17	(Purwanto et al., 2023)	Qualitative interview model	Organizational learning, ambidexterity, adaptability	Corporate context	Thematic analysis	Conceptual findings validated qualitatively
18	(Ibrahim et al., 2025)	Questionnaire	Grit, burnout management, learning persistence	Students	Regression analysis	Significant effect found

Table 3 (continued). Data Extraction of Learning Agility Measurement Across Selected Studies

No	Author & Year	Instrument/ Model	Dimensions of Learning Agility	Sample/ Context	Analytic al Method	Psychomet-ric Findings
19	(Das & Rajini, 2024)	360-degree feedback instrument	Leadership development, adaptability, performance growth	Managers	Delphi & survey analysis	Consensus achieved
20	(Şimşek et al., 2025)	Survey questionnaire	Learning motivation, self-regulation, adaptability	Students	SEM	Model fit confirmed
21	(Vatsa et al., 2025)	System dynamics model	Achievement motivation, entrepreneurial adaptability	Students	System modeling	Simulation results validated
22	(Park et al., 2022)	Survey instrument	Career resilience, employability, adaptability	Employees	SEM	Significant relationships
23	(Aliyyah et al., 2024)	PLS-SEM questionnaire	Employee agility, organizational agility, adaptability	Company employees	PLS-SEM	Model supported
24	(Lubis et al., 2025)	Survey instrument	Collaboration, organizational behavior, synergy	Educators	SEM	Statistical significance confirmed

Discussion

RQ: How is learning agility measured in existing empirical and conceptual studies, and which models, dimensions, and measurement approaches are most commonly used?

The findings of this review directly address the research question by showing that learning agility is predominantly measured using Likert-scale-based self-report instruments, most commonly operationalized through multidimensional models consisting of cognitive, people, change, and results agility. Based on the data extraction table, the most frequently used analytical approaches are Structural Equation Modeling (SEM) and PLS-SEM, while instrument development tends to rely more on adaptation of existing scales rather than the development of new measurement models. These patterns indicate that the learning agility measurement is relatively standardized in structure but varies in depth and contextual application.

Various studies on learning agility measurement demonstrate relatively consistent methodological tendencies, although each study emphasizes different contexts, resulting in an increasingly diverse conceptual landscape. Across educational, healthcare, and organizational settings, learning agility is predominantly measured using Likert-scale-based instruments supported by psychometric validation. However, the depth of operationalization varies significantly among these studies. Several studies, such as [Boulant-van Dam et al. \(2022\)](#), [Ab Jalil et al. \(2022\)](#), and [Jung & Lee \(2024\)](#), reflect a more rigorous instrument development process through comprehensive applications of exploratory and confirmatory factor analysis (EFA and CFA). In educational contexts, studies such as [Lubis et al. \(2025\)](#) highlight the role of learning agility in curriculum effectiveness and educator performance, while [Huang & Kou \(2025\)](#) validate its contribution to student resilience and well-being through a mixed-methods approach.

In contrast, other studies, including [Wolor et al. \(2025\)](#), [Narajah et al. \(2025\)](#), and [Jian \(2022\)](#) tend to rely on the adaptation of existing instruments without introducing new indicators. This suggests that learning agility measurement remains largely dependent on static self-report approaches [Li & Chalermvongsavej \(2025\)](#), relying heavily on subjective perceptions. Although mixed-methods approaches offer opportunities to enrich measurement practices, their application remains limited, with the minimal use of methodological triangulation or diverse data sources. Furthermore, research by [Mahmutoğlu et al. \(2025\)](#) shows that learning agility is not only measured through individual perceptions but also functions as a mediator in the dynamics of school leadership's influence on teachers' organizational commitment, demonstrating the applicability of learning agility measurement in the context of educational management.

Differences in the depth of operationalization also influence variations in learning agility dimensions across studies. While cognitive, people, change, and results agility consistently emerge as core dimensions, their relative importance varies depending on the context. In higher education, studies such as [Patwardhan et al. \(2022\)](#), [Jian \(2022\)](#), and [Ab Jalil et al. \(2022\)](#), emphasize cognitive agility as a key predictor of student readiness. Similarly, [Konstantinou & Miller \(2021\)](#) highlight learning agility as a bridging competency between academic learning and workplace experience. In contrast, organizational studies [Carmeli et al., 2021](#); [Purwanto et al., 2023](#); [Tripathi & Kalia, 2024](#) prioritize change agility as a critical driver of innovation and adaptability, particularly in response to digital transformation. Moreover [Şimşek et al. \(2025\)](#), position learning agility as a determinant of perceived ease of use in generative AI adoption, reinforcing its relevance in technology-driven environments.

From a social and interpersonal perspective, studies by [Narajah et al. \(2025\)](#) [Jo & Hong \(2022\)](#), and [Wolor et al. \(2025\)](#) identify people agility as the most critical dimension in highly collaborative settings. [Das & Rajini \(2024\)](#) further extend this perspective by positioning learning agility as an indicator of leadership intervention effectiveness through 360-degree feedback, highlighting its role in talent and leadership development. These findings confirm that, although the structural dimensions of learning agility appear consistent, their relative importance is context-dependent, as reflected in the variation in dimensions reported in the data extraction table. Nevertheless, the literature still lacks consensus on the prioritization of these dimensions, indicating the need for further comparative research across contexts.

The analytical approaches employed in these studies revealed both robust patterns and notable limitations. Structural Equation Modeling (SEM) and Partial Least Squares SEM (PLS-SEM) dominate the analysis, as seen in studies such as [Ibrahim et al. \(2025\)](#), [Lubis et al. \(2025\)](#), [Park et al., \(2022\)](#), [Tripathi & Kalia \(2024\)](#); [Vatsa et al. \(2025\)](#), which treat learning agility as a complex latent construct. However, this dominance often leads to learning agility being positioned primarily as a mediating variable, with limited exploration of its underlying psychological mechanisms and antecedents. Additionally, the use of bibliometric analysis and topic modeling [Grau García et al. \(2025\)](#) [Zykiene et al. \(2021\)](#); suggests a shift toward digital organizational contexts, although these approaches have not yet addressed the need for longitudinal designs or behavior-based measurement. [Aliyyah et al. \(2024\)](#) contribute by demonstrating that various forms of agility, including learning agility, play a significant role in increasing company productivity and profitability, strengthening the strategic relevance of learning agility to organizational performance.

Theoretically, various studies show that learning agility is not only about the ability to learn quickly but also a set of adaptive capacities formed through a combination of experience, reflection, motivation, and interaction with the social environment. Research in education [Li & Chalermvongsavej \(2025\)](#), [Ab Jalil et al. \(2022\)](#), [Jung & Lee, \(2024\)](#), [\(Bouland-van Dam et al., 2022\)](#), and change management [Carmeli et al. \(2021\)](#); [Purwanto et al. \(2023\)](#) has generally shown that learning agility functions as a psychological mechanism that bridges various internal resources such as self-leadership, self-confidence, and resilience with various outcomes including performance, innovation, professional commitment, and readiness to face uncertain conditions. However, most studies still place learning agility at the individual level, even though findings from [\(Carmeli et al., 2021\)](#) indicate that learning agility can also operate in the inter-organizational realm. This suggests

that the conceptual scope of learning agility can be expanded beyond the individual, although theoretical research in this area is still very limited.

Overall, this review demonstrates that although learning agility measurement has a strong psychometric foundation, several critical limitations persist. These include an overreliance on self-report instruments, limited use of diverse and multi-source data, a lack of longitudinal research designs, and insufficient development of behavior-based measures. Although modern psychometric techniques enhance reliability, they are not yet fully capable of capturing the dynamic and evolving nature of this concept.

In technology-driven environments, these findings highlight the importance of integrating learning agility measurements with digital learning systems. The use of Learning Management Systems (LMS), adaptive learning technologies, and artificial intelligence (AI)-based educational tools enables a more dynamic and real-time assessment of learning behaviors. Such integration has the potential to overcome the limitations of traditional self-report instruments and provide more objective and context-sensitive insights into individuals' learning agility, particularly in digital and technology-enhanced learning environments.

CONCLUSION

This systematic literature review indicates that learning agility is predominantly measured using self-report instruments based on multidimensional models, particularly the four core dimensions of cognitive, people, change, and results agility, with psychometric validation techniques such as exploratory and confirmatory factor analyses (EFA and CFA) and analytical approaches largely dominated by SEM and PLS-SEM. While these findings suggest a relatively consistent measurement pattern across studies, they should be interpreted with caution due to several limitations, including the strong reliance on single-source self-report data, the limited use of behavioral and longitudinal approaches, and variations in construct operationalization across contexts. In addition, this review is constrained by methodological limitations, such as the inclusion of only accessible full-text articles, potential publication bias, and reliance on abstract-based data extraction, which may restrict the depth of the analysis. In technology-driven learning environments, these limitations highlight the need for more adaptive and integrated measurement approaches, particularly through the use of digital learning systems such as Learning Management Systems (LMS), adaptive technologies, and artificial intelligence (AI)-based tools that enable more dynamic and real-time assessment. Therefore, future research should focus on developing context-sensitive, multi-source, and longitudinal measurement models to better capture learning agility as a continuous and evolving process, rather than a static individual capability.

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AUTHOR CONTRIBUTION STATEMENT

NM: conceptualization, methodology, data curation, formal analysis, writing – original draft preparation.

MK: validation, visualization (PRISMA diagram), review and editing, and supervision.

MS: Investigation, resources, data validation, writing – review and editing.

AI DISCLOSURE STATEMENT

The author used ChatGPT during the preparation of this work for language refinement and structuring of the manuscript. After using the tool, the author thoroughly reviewed and edited the content as needed and takes full responsibility for the content of the publication.

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